MLPC – Team Toothpaste

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Outline

Problem

- Better accuracy isn’t everything
- Missing event (false negative) penalized more than False alarm (false positive)

Our Goal

- Design cost-aware system

3 Hypotheses

- “Dumb” threshold is costly
- Raw predictions are noisy
- Deeper isn’t always better

Results & Conclusion

Class	TP	FP	TN	FN
Speech	0	1	0	5
Dog Bark	0	1	0	5
Rooster Crow	0	1	0	5
Shout	0	2	0	10
Lawn Mower	0	3	0	15
Chainsaw	0	3	0	15
Jackhammer	0	3	0	15
Power Drill	0	3	0	15
Horn Honk	0	3	0	15
Siren	0	3	0	15

Table 1: Cost Matrix for Evaluation

Architecture

Multi-stage pipeline

1. **Input:** Frame-level audio embeddings (120ms)
2. **Core Model:** Bidirectional GRU classifier for temporal patterns
3. **Post-Processing:**
    1. Smooths probabilities
    2. Applies tuned thresholds
    3. Filters out noise
4. **Output:** Final predictions for 1.2s segments

Hypothesis 1: “Dumb” Threshold is Expensive

Hypothesis

- Single fixed threshold (e.g., 0.5) performs poorly
- Ignores unique cost penalties

Method

- Tuned one threshold for each of the 10 classes
    - Iterated through all possible thresholds
    - Selected the one with lowest cost for that class
- Single most impactful improvement

Class	Threshold
Speech	0.36
Shout	0.39
Chainsaw	0.33
Jackhammer	0.55
Lawn Mower	0.50
Power Drill	0.54
Dog Bark	0.29
Rooster Crow	0.49
Horn Honk	0.59
Siren	0.43

Table 2: Optimal threshold for each class based on cost

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Hypothesis 2: Raw Predictions are Noisy

Hypothesis

- Model output contains short, spurious detections -> expensive
- Need to clean these predictions

Method

- Two-step “cleaning” process on frame-level predictions:
    1. Median Filtering: Applied to raw probabilities to smooth out single frame “blips” and fill small gaps
    2. Duration Filtering: After thresholding, remove any detected event shorter than 3 frames (360ms)
- Effectively removed noise -> lowering final cost

Hypothesis 3: Deeper Isn’t Always Better

Hypothesis:

More complex model might overfit -> higher cost

Method

- Trained models with varying complexity:
  - **Layers:** [1, 2, 4]
  - **Hidden Dimensions:** [256, 512, 1024]
- Compared validation/total cost after full pipeline

Outcome

- 1 layer & 256 hidden dimensions achieved lowest cost

Results & Conclusion

Key Strategies:

- Cost-sensitive loss function
- Class-specific probability thresholds
- Median and duration filtering

Performance

System	Total Cost on Test Set
Baseline (Most Frequent)	101.47
Logistic Regression	53.69
Default RNN	38.22
Regular RNN (tuned parameters)	37.92
Tuned RNN	35.95

Table 3: Final performance of each classifier